8

Actuators

At this point in the automation machine design process, one might think that the remaining steps would simply be filling in the details. Find the right size of motor and be done with it. Let someone else do the dirty work and move on to the next challenge, but this could not be further from the truth. Finding the right type and properly sized actuator (a device that produces motion with some force or torque) will often return a designer to an earlier step in the design process, sometimes all the way back to the original concept.

The human muscle is one of the most compact and efficient actuators ever built. Trying to emulate this wonderful design will cause one to be frustrated. Perhaps this can best understood by noting that there are currently no prosthetic devices available to replace a missing arm or leg that have self-contained power and actuation. In such an application, if one could achieve this result, it would seem to be more cost justifiable than its application in a generic automation device.

8.1. TYPES OF ACTUATORS

If one looks past the human muscle and focuses on the types of currently available actuators, one will generally see a list something like this:

- compressed air (pneumatics);
- hydraulics;
- electric motors.

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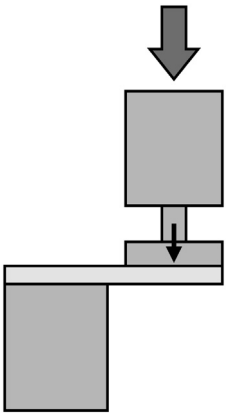


FIG. 8.1 Machine using linear output

There are subcategories to each of these actuator types, and there are some newer developments that might not seem to fit, but these types of devices where power of some sort is used to generate a motion with a force or torque are the building blocks for automation.

As an example, let us look at a simple application. In Fig. 8.1, there is a metal plate that is originally flat. Ultimately, in Fig. 8.2, the plate is bent. The questions needing to be addressed include:

- How thick is the plate?
- What is the plate's material composition?

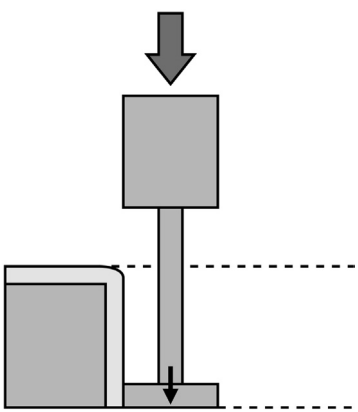


FIG. 8.2 Machine completing operation

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- What is the bending force?
- Is there lubrication used to assist the bending process?
- How fast does the bending need to occur?
- Will the plate have springback?
- How many plates will be bent on this machine?
- What are the currently available power sources?
- Where will this bending machine be located?

Each of these questions will help us to understand the challenges of actuator selection.

The plate's thickness, composition, and bending force are all related. For most materials in use today there are handbooks that will assist the designer to determine these relationships. For a given thickness and material type, the information to calculate the bending force will be provided. However, if it is a newer composite material, it is possible that the designer will have to perform some tests to determine these relationships. So a known process is less risky than an unknown process, to no-one's surprise.

Other answers to these first three questions might produce other, more obscure, concerns. If the plate material had carbon fibers internally, and if the plate could be bent successfully, would the process release some carbon fiber particles in the working environment? Would these carbon fibers mean that the actuator needs to be mechanically shielded from the surroundings? This is not a big deal for hydraulics or pneumatics, since the shielding option is readily available, but what if the electric motor uses air flow for cooling? The surrounding air, if it contains carbon fibers, could create havoc with the electrical contacts. Perhaps the contacts become dirty very quickly, or, on the other end of the spectrum, the contacts get shorted out.

Does the bending process need lubrication? Will the lubrication cause the actuator to fail in a few weeks or months? Will the machine need to be cleaned periodically? Or will the actuator leak hydraulic fluid onto the plate, and thus the work area become contaminated?

The length of time allowed to bend the plate will greatly affect the sizing of the actuator. If a gearbox can be used, and assuming that gearbox friction does not become excessive (and in real life this cannot be ignored), one could power this plate bending press with a squirrel in a cage. Most likely the squirrel would have to run for hours to do this task, similar to the length of time an analog wind-up clock takes to move the hour hand. It can be forceful, but not practical in industry.

Springback occurs with most metal plates. Even though the metal chosen is not technically to be used as a spring, it will recover from the pure 90° bend as shown in Fig. 8.2 to perhaps 89° or at least 89.99°. This springback

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issue is a concern for the manufacturing engineer who wants to use this machine, but it also can create issues to the automation engineer. Will there be a frictional force to return the moving bending fixture to the original home position? What will the force level be, and can the process be done quickly so as not to waste time?

Imagine if the squirrel cage method worked (Fig. 8.3), and imaging that the springback forces were minimal. But the squirrel was made to turn around and rotate the cage in the opposite direction. Now it would take about the same amount of time for the moving bending fixture to be raised to the home position. The gearbox ratio would possibly be excessive for the returning motion, assuming that the bending fixture was not extremely heavy. Could one shift gears like one does for a 10-speed bicycle or a manual transmission automobile to make the return motion happen quicker?

So in real life (without the squirrel), the actuator requirements might be such that the return to home motion be a different mode than the bending motion. It should return faster and hopefully with lower loading requirements, but maybe not. Changing gear ratios is not cheap, so other machine design options might need to be explored for the optimal costing and performance cycle time.

The question concerning how many plates are to be bent addresses the life expectancy of the automation machine. This is parallel to you as a home owner purchasing an electric drill at a lower price than that of a commercial grade drill. As a homeowner, you might use the drill 10 minutes to a few hours per year, depending on your skills and your (or your spouse's) ambitions. You expect the drill to last for a while, but if during a large construction project it

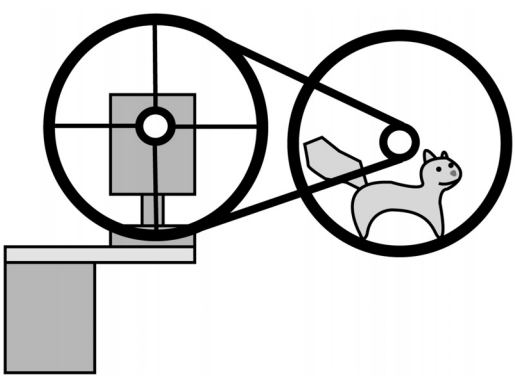


FIG. 8.3 Low power source geared to generate a large but slow force

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up and dies, you purchase another one without much disappointment. It was not too expensive in the first place. However, if you are a contractor drilling 8 to 12 hours per day, you spend double or triple the amount to have a drill that you expect to last for four or five years of heavy use. Your drill cannot let you down when there are deadlines or you have storms approaching and the roof is not attached yet.

Similarly, for the bending of plates, is this a homeowner's level of use, or 24/7 operations? There are both actuators that can handle the high level of use, and those from manufacturers who seem to specialize on inexpensive (we will not use the word cheap here, though perhaps appropriate) actuators for a machine that is not destined for longevity. Most actuators come with some type of duty cycle data, which is invaluable in making this judgment. Some customers will specify the machine life expectancy, which will dictate the performance level of the actuators.

The question about available power sources can be huge. Many large electric motors used in larger machines are 240 V AC three-phase power. For many industrial manufacturing buildings this is already available, but if the building is in a rural locale, or if the area is solely residential, three-phase power can be very costly to install. And as a machine builder, one has a limited potential in the horsepower output of 240 V single-phase motors.

Pneumatics (compressed air) can be seen as a limitless supply of energy if a factory is already outfitted. But the issue of capacity can sneak up and surprise you. The current number of machines might be maxing out the compressed air system. Your new machine might be the sole reason an upgrade or improvement is required, and your machine might need to bear the entire costs. The next machine may get a free ride on your machine's back.

Hydraulic power supplies are usually separate for each machine they power. Rarely will there be a plantwide hydraulic power supply available to tap into. These supplies are often located near the equipment they power. However, the accumulated noise of many such powered machines might become excessive upon the implementation of your machine. Although chances are operators would be using ear protection before your machine gets installed, it could be the straw that breaks the camel's back for noise pollution.

Related to this is the final question on our list, the machine's location. If the machine is handling unpackaged food products, it usually must get washed down once each day. This can be a big issue in deciding to have waterproof electric motors. Or it can limit the hydraulic actuators to not having any drips of fluid that can get into the food. Even pneumatics, which will often vent out the air from one end of the cylinder during motion, must be filtered to keep out oil droplets since many compressed air systems deliberately add oil particles to keep the cylinder's internal systems from rusting out.

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8.2. APPLICATION CONCERNS

The number of different actuator applications is numerous, but there are a few issues that bear mentioning in general:

- Amount of work to be done;
- Frictional losses;
- Dispensing viscous materials;
- Gearing and inertial;
- Conveyors;
- Intermittent motion.

These each represent a larger list of possible concerns that make actuator selection more difficult than one would expect.

8.2.1. Amount of Work to be Done: Power = Force × Velocity + Friction

The basic amount of power a machine needs to provide can be calculated idealistically. If one returns to the machining processes, one can look at the anticipated force and multiply it by the velocity and have a determination for power. If it is a rotational situation (Fig. 8.4) where metal is being removed, one needs to determine the anticipated torque and the rotational velocity to calculate the power.

If the automation concerned is not a machining process, then one needs to look at the motions and weights or inertias to determine the idealistic power requirements, but just like the problems in high school physics, it supposedly takes no work for one to walk on a level floor. In reality, there are always losses to account for, and therefore one needs actuators larger than the physics world states.

8.2.2. Frictional Losses

Friction is normally the machine designer's enemy. Except when one wants friction to assist with bringing a machine to rest quickly, friction adds to actuator size

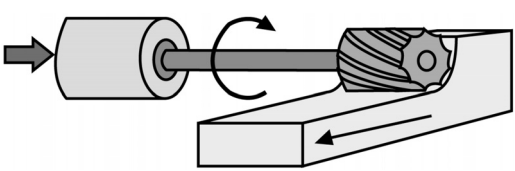


FIG. 8.4 Rotary output power

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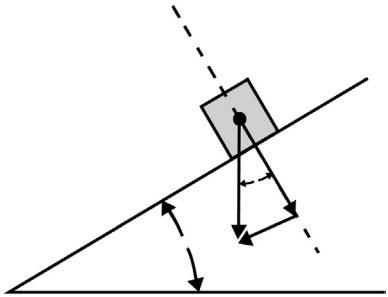


FIG. 8.5 Static friction

and power consumptions. Many machine components need to overcome both static friction (Fig. 8.5) and kinetic friction (Fig. 8.6). The proper use of bushings and bearings will hopefully keep fiction minimized as much as possible, and prevent a situation where there is significantly larger static friction compared to kinetic friction occurring. It is difficult to anticipate the resulting motion when high static friction finally gives way to the lower kinetic friction, particularly when the static friction may change over time, temperature, and number of cycles. If this motion is to happen in a predetermined time sequence, it can be troublesome.

There are also process situations where an actuator needs to overcome a viscous frictional situation. This can appear in many configurations, but two common forms are linear (Fig. 8.7) and rotary (Fig. 8.8). The rotary form can also be seen in journal or sleeve type bearings, but is hopefully less significant than the friction one sees in a process such as Fig. 8.8.

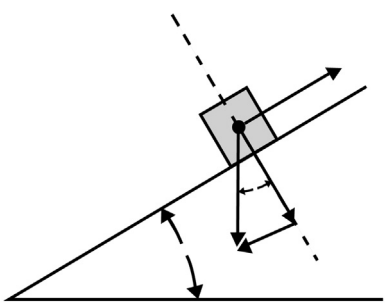


FIG. 8.6 Kinetic friction

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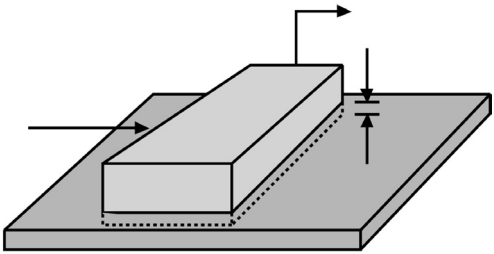


FIG. 8.7 Viscous friction

8.2.3. Dispensing Frictional Materials

Some frictional situations cannot be avoided. And in some cases, one does not want to avoid it, but wishes to have it under control in a known environmental condition. The dispensing of a viscous material (Fig. 8.9), such as an adhesive or a silicon product, cannot significantly modify the liquid's frictional properties. If the material was too runny, it might drip from the dispensing head. Or if it freezes up when the plant's interior is no longer heated to 65° during the winter months as a cost-saving effort, the entire automated process may have changed so as to threaten quality or production rates.

8.2.4. Gearing and Inertia

There are many textbooks available to assist one in calculating the rotation moments of inertia. And if one has infinite time to conduct a project design, one can calculate the effect of every tapped hole, screw body and head, and length of sensor wire. As stated in Chapter 6, these machines are most like "not going to

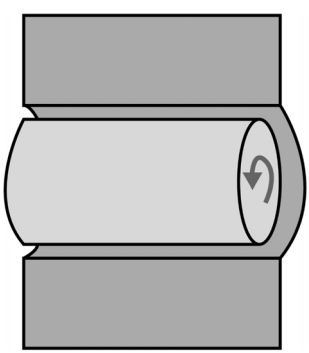


FIG. 8.8 Rotary viscous friction

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FIG. 8.9 Dispensing viscous materials

fly," and the prudent thing to do is to calculate the significant contributions and add in a slight factor of safety to cover the minor details. Now if a CAD model will produce the entire results from one mouse click, then do not ignore the results, but student teams have gone through the erroneous effort of determining inertia contributions down to the 0.1% level, only to add or subtract some component two weeks later in the design.

There are two common approaches to inertia calculations when one designs a machine. When there is a gearbox involved (Fig. 8.10), one will want to know

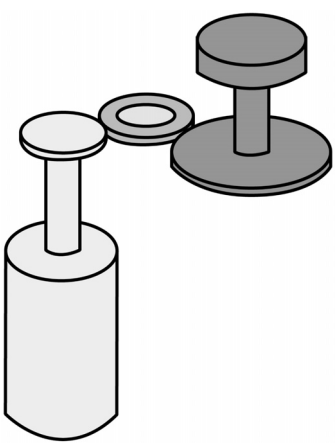


FIG. 8.10 Gear drive application

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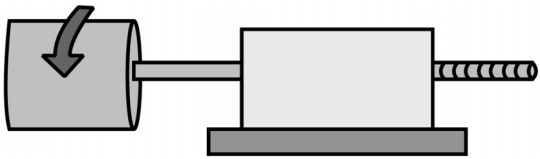


FIG. 8.11 Rotary to linear drive application

what the impact the load shown on the right will have on the motor on the left. The gear ratio is used to calculate the reflected load seen by the motor for sizing considerations. This is analogous to calculating the torque one produces by pressing the foot downward on a bicycle petal compared to what the torque is at the rear wheel. In order to climb a hill at a desired speed, it depends on what gear has been engaged for the required effort.

For many motion applications in automation, a rotational motor will use a threaded shaft to move the load in a linear fashion (Fig. 8.11). So for a target linear speed and force, the pitch of the shaft and the block will determine the torque and speed required of the motor. If one has all the time in the world to linearly move the load, the shaft pitch can be very fine, and one can simply use a rechargeable screwdriver to power it, but most automation applications will never be satisfied with this approach.

8.2.5. Conveyors

Conveyors at first glance do not look to be anything special for actuator sizing purposes. There is a motor usually connected to one roller (Fig. 8.12) in one of several arrangements. More commonly, the motor is connected via a right angle gearbox, so as to match a reasonably sized motor to the needed belt speed, and to take up less floor space, but a conveyor may see a significantly varying loading depending on the application. If the conveyor is to be used to load luggage into an airplane cargo space, it may be started in motion with no suitcases on it. Then the loading will increase as baggage handlers toss on your hopefully nonfragile suitcase. However, if there is trouble unloading it at the top, the conveyor may be stopped abruptly for safety reasons, and then abruptly restarted.

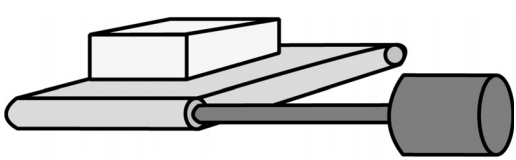


FIG. 8.12 Conveyor drive application

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Whether or not a conveyor in its application is allowed to gracefully accelerate the belt and load up to speed (and gracefully decelerate to a stop) has great motor sizing impact compared to needing the belt to immediately resume top speed or to come to a stop in a fraction of a second. Nothing is more frustrating than a piece of automation built around a slower accelerating conveyor, and then having the customer update their requirements to having it instantly start at 60 miles per hour.

8.2.6. Intermittent Motion

Some requirements in automation have dependable intermittent motion. Dependable and repeatable (hopefully like your heartbeat) so that having an actuator go on and off all of the time makes a machine designer look to older proven technology. Dial Machines (or Rotary Tables) can be used to index 24 hours a day, and therefore have been using a constant speed motor to drive the input (right) side of the Geneva mechanism in Fig. 8.13. The pin on the disk goes into the slot and rotates the six part form on the left in 60° increments. The normal geometric motions cause some relatively smooth accelerations and decelerations, but there is still a variable loading occurring to the drive motor. The cyclical application of the increased and then decreased inertia almost begs for a flywheel to even out the required loads. This can be beneficial to a limit, until the flywheel adds so much moving mass that the system almost must continue to the next motion increment rather than stop on command.

8.3. PNEUMATICS

Pneumatics is the use of compressed air through pipes and hoses used to move something. Most times the compressed air goes into a rigid device that is capable

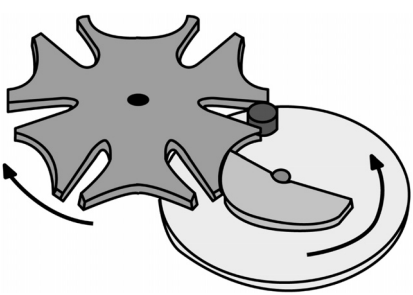


FIG. 8.13 Rotary table with Geneva mechanism

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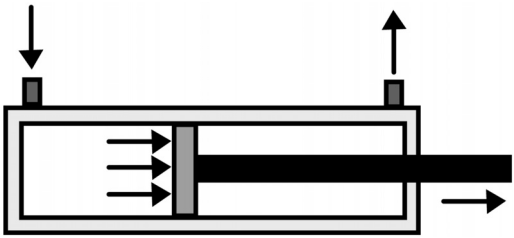


FIG. 8.14 Pneumatic air cylinder: force = pressure × area

of a defined motion. The great majority of these devices can be modeled by Fig. 8.14. Here, air enters though the hose fitting on the upper left of the cylinder and fills the left chamber. The increased volume of air moves the piston (and rod) to the right. The air in the right chamber is normally allowed to exit through the hose fitting on the cylinder's upper right. If the air in the right chamber is not allowed to exhaust, it will become compressed. Then the resulting motion will be limited to some proportion related to the difference in pressures on each side. Rarely is this used in practice.

Since air is quite compressible, the most easily achieved repeatable motion is to allow enough air to go into one side (or the other) so as to drive the piston to one end. Then the speed and motion of how it got to the end is hopefully not critical. If the desired motion has very specific velocities and/or accelerations, it is probably not best to use pneumatics.

There are hundreds of types, models, and variations to the generic air cylinder shown. A few of them, represented by their stick figure models, are shown in Fig. 8.15. Almost all of these are designed to work at standard shop air pressure of 80–90 psi. Air cylinders do wear out from frictional issues and leaky seals around the piston rod. Contaminated air can induce particles that can cause havoc to the cylinders' moving surfaces and the control valves, so filters on the air line are a must.

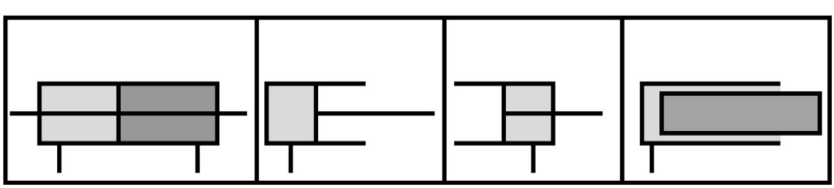


FIG. 8.15 Pneumatic cylinder types: (a) rod on each end; (b) single direction rod external support; (c) single direction rod internal support; (d) piston serving as the rod

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Major types include:

- Single action direction (Fig. 8.15b and c);
- Single action direction with spring return;
- Dual action (Fig. 8.14);
- Pancake style: fat but very short with limited stroke;
- Telescoping: multistage, similar in design to multiple part TV antenna;
- Rotary;
- Multiple position: other small air cylinders stop piston from traveling to end

Other options include:

- Location of hose fittings;
- Cylinder mounting hardware;
- Integrated sensors to confirm travel is complete;
- Internal cushions to dampen piston deceleration;
- External shock absorbers to dampen piston deceleration.

To make the cylinder in Fig. 8.14 reverse direction, the hose connections need to be changed. Manual valves can be used to switch which cylinder end is connected to the compressed air source, but using manual valves is not practical for automation. To have a controller operate a pneumatic cylinder, electrically signaled valves are used. Most of these use an electromagnetic solenoid to move a small piston opening or closing one or more ports. The machine builder is responsible to configure the hoses to the correct ports to get the desired performance. In most applications, pneumatics is more convenient than hydraulics since there is no effort to recover the exhausted air, so the hose and valve configuration does not have to be as complex.

The size and shaping of all of the hoses and fittings will have a great impact on performance (Lentz, 1985). Figure 8.16 shows the general trends of most pneumatic equipment. As one moves away from the origin, each curve represents a larger valve diameter. The impact of both air lines and valves of insufficient diameter is sometimes hard to predict. It is not uncommon in developing totally new automation approaches that the entire pneumatic system needs to be upgraded to a larger diameter to get satisfactory performance.

Pneumatics components are available in English (inches) and metric (mm), and various commonly used sizes are so close to one another (and sometimes not properly labeled) that a pile of leftover parts is difficult to distinguish. A cross-threaded metric part into an English fitting can cause a world of problems. Another issue is that the working area of the piston is not the same on both sides due to the rod's connection to the piston, so the motion may not be the same in both directions for the same air supply.

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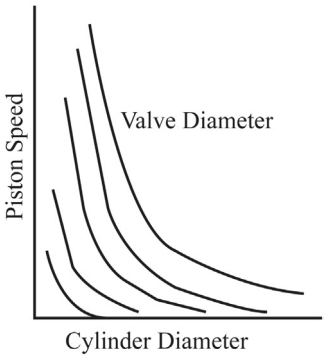


FIG. 8.16 Cylinder diameter vs. piston speed with respect to valve diameter

8.3.1. Advanced Pneumatics Devices

One design issue is the fact that the generic cylinder shown has the rod sticking out. The overall dimension in the direction of motion is a little more than twice the piston motion. Sometimes a designer cannot fit the cylinder and the rod into the machine envelope, so an alternative design is the rodless air cylinder. One common method is to remove the rod and attach a strong permanent magnet to the piston, and add to the cylinder an external sliding magnetic piece that follows the inside piston magnet. As long as the applied loading is less than the magnetic attraction, it will stay coupled. There are more than a dozen other rodless cylinder designs available, many using tricks other than magnetism.

Another recent pneumatic device is the air cylinder with servo control. For many years this was but a dream, since air is highly compressible compared to a liquid. If one was to use a manual valve and try to keep a cylinder piston at some point between its limits by flipping the switch back and forth, the results were very crude. Even electronically, the results were within an inch or two at best. However, recent research has helped industry to produce commercially available air cylinders with great servo control. A dedicated microprocessor monitors a continuous position sensor to determine the current piston position. Then the microprocessor switches the input source back and forth at a tremendous rate, also understanding the compressibility effects of air and the time delays in the motion response, and the result is highly accurate. This product is a true outcome of Mechatronics research.

8.3.2. Pneumatics Vacuum Generators

A valuable subset of the pneumatics toolbox is the method of using compressed air to generate a vacuum. Normal vacuum pumps driven by electric motors are

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available, but they are in general quite substantial in size. If the vacuum pump is not closely located to the suction cup, there is also a time delay while the connecting hose draws down to the vacuum level. If one needs a vacuum cup to grab something in a tight environment, or if payload restrictions eliminate the possibility of a vacuum pump, a Venturi vacuum generator (Fig. 8.17) can create an unbelievable amount of suction in a very short amount of time. There are even dual and triple stage units that can produce low pressures and reasonably high flow rates.

The Venturi device has compressed at flowing through from the left-hand side. The restriction in flow near the middle of the device creates a pressure drop (this is perhaps the only useful item found in a fluids course textbook that automation engineers can use to their advantage). There are only two real limitations to this device. The first is that they take a constant supply of air pressure flowing through them to hold an item. This is different to an air cylinder, which does require continued pressure to keep something in place, but not the continued flow rate of a Venturi device. Secondly, if the environment of use is dirty, the device will pick up particles. There is nothing as frustrating as cleaning out instant cocoa drink powder from a Venturi device when some of the transported pouches break apart.

If an automation application requires multiple suction cups, and there is even the slightest chance that one of the cups might not get a good seal, so therefore would not achieve a vacuum, it is common to see every one or pair of suction cups having their own Venturi generator. The incremental cost is not unreasonable, particularly compared to losing the grip on a costly transferred product.

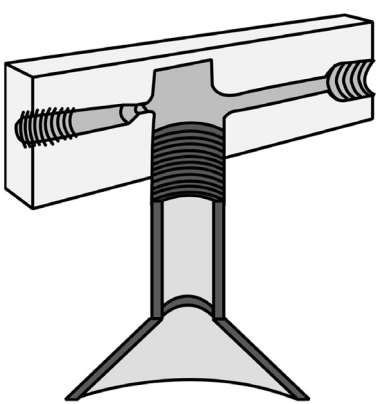


FIG. 8.17 Vacuum generator and suction cup: cross-section

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Pneumatics systems are relatively cheap to purchase, cheap to install, and cheap to run. Some machine builders swear by them, others, because of nasty operating conditions, swear at them. Some builders will never use them if given a choice, but by having the power source (the air compressor) at some other location, there can be a great deal of action happening in a compact working volume.

8.4. HYDRAULICS

Hydraulics is like the big brother to pneumatics, a much stronger, noisier, more dangerous, and potentially messier big brother. The basic concept of the air cylinder in Fig. 8.14 stays the same, but some of the operating conditions change tremendously. Hydraulics use a fluid similar to your car's power steering system rather than air, so compressibility is orders of magnitude less. Many hydraulic systems operate in the 2000–3000 psi range, so the cylinder, hoses, valves, and power sources all need to be upgraded. One could never use a pneumatic device for a hydraulic application. A leak in a hydraulic hose or fitting can produce a high-pressure stream that acts like an abrasive waterjet cutter, able to slice off one's fingers and the like.

Figure 8.18 shows the traditional components in a minimal hydraulic system. The fluid starts out in the reservoir. The pump brings the fluid up to the 2000–3000 psi pressure and it fills an accumulator. The accumulator's role is to have a large supply of high-pressure fluid available for short bursts that might normally outstrip the pump's capacity. It acts analogously to a capacitor in an electrical circuit. However, the pump runs continuously (a real noise source), and if the accumulator and the hydraulic cylinders are not in need of the fluid, a valve between the pump and accumulator dumps the excess fluid back into the reservoir. So the energy consumption of the system is always on full throttle as soon as the pump is turned on.

Any thoughts of turning the pump on only when one needs the pressurized fluid are dashed by the requirements of the systems. The fluid needs to be warmed up and any bubbles removed. Bubbles of air are compressible, and would compromise the overall performance. This has been one of the limitations of hydraulics systems.

A system of valves is used to control a hydraulic piston as shown in Fig. 8.18. Since the supply pressure is constant, the larger the cylinder, the larger the force. In this simple representation, the valve will only allow the piston to move to the left. The fluid in the cylinder's left chamber will be pushed back into the reservoir. This, in practice, is quite useless. There needs to be a more complex valve setup that allows the cylinder to move in both directions. This is achieved by directing the pressurized fluid into either side, and allowing the returning fluid to route to the reservoir.

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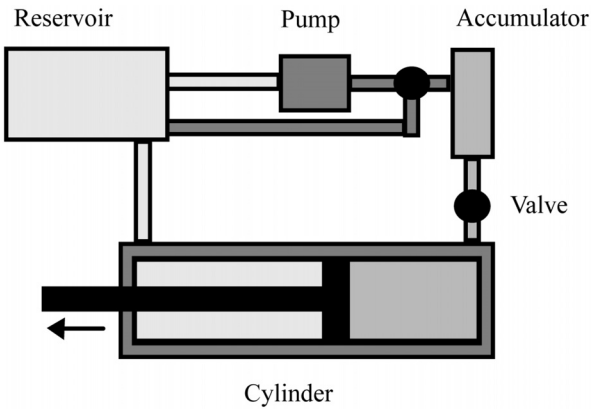


FIG. 8.18 Hydraulic actuator system: force = pressure × area

Besides being noisy, hydraulics is often messy. When there is an oily fluid being forced at 3000 psi through fittings and held in by flexible seals, one can imaging that there would be leaks, either visible drips or a smelly film in the air. One only has to look at the grease (and the dirt it attracts) found on most backhoes or other construction equipment to see the potential concerns. One would not like to see a backhoe used in the baking of cookies, for example, but in high-force applications like metal working presses, there are no logical alternatives.

The early development of industrial robot arms was totally based upon hydraulics. This was chiefly because the electric motor industry was nowhere near where it is today. Early robots, having hydraulic pumps and reservoirs, were used to lift heavy metal parts into such applications as foundries and the welding of car bodies. In many situations, this was seen as a great advance at the time, and the leaks were seen as something one had to live with.

An interesting automation development by IBM in the late 1970s and early 1980s was their RS/1 robot. It was a gantry style robot that used a novel method of cycling multiple rollers over a machined metal repeating sine wave. The multiple rollers were actuated by four small hydraulic cylinders, similar in size and configuration to the valves found on a trumpet. By energizing the four valves in a definite pattern, motion in one direction or the reverse could be achieved. It was a technological marvel of the time. However, it was a robot designed to assemble electronics that was hydraulically driven. It gave off an invisible yet detectable (by the human nose and fingers) film that would coat everything in the working environment. Experiments at RPI by the author showed that it took only several hours of operation for the friction coefficient between the robot's gripper fingers

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and a block of aluminum to change from 0.4 to 0.1. The great technology advancement that it represented was limited by its type of power source. IBM then converted the next generation RS robot to use electric motors.

8.4.1. Hydraulics in Automation Today

Except where hydraulics is still needed to generate a great force with a remote power source (the reservoir, pump, and accumulator do not need to be directly next to the actuator), the number of automation applications using hydraulics has dropped. There are almost no remaining robot designs using hydraulics, and many older packaging machines have been upgraded. Electric motors have evolved to become far more efficient and compact than they were 20–30 years ago, and they have made a huge inroad into automation.

8.5. ELECTRIC MOTORS

The basic DC electric motor has been around for a very long time. It is still used for some applications, but has often been replaced by newer technology. The basic DC motor as perhaps you wound in a high school technology class uses a stator (Fig. 8.19) and has windings of lacquer-coated insulated wire (Fig. 8.20).

The motion is generated by the timed changing of the stator field, as shown in Fig. 8.21. Depending on different motor parameters, speeds could be changed by varying the applied voltage.

This kind of motor is useful if one is designing a printing press for making the morning paper. Motors and the attached hardware can come up to speed gradually or quickly, always producing a newspaper per cycle, no matter what

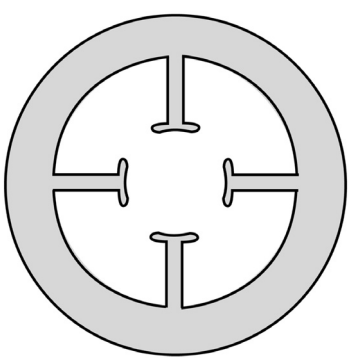


FIG. 8.19 Electric motor stator laminator

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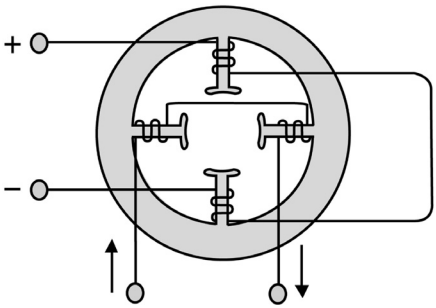


FIG. 8.20 Four-pole stator

the speed, but for something like a robot arm, where one wants the robot to go to a programmed position and stay there, a standard motor is not too useful.

8.5.1. Electric Servomotor

So the servomotor was developed, where the motor was designed to rotate to a fixed angular position and remain there. This is something one cannot achieve with a normal electric motor found in a power tool such as an electric drill. This process relies on a sensor located on the motor output shaft to report what the current angular position is. A comparator in the motor controller (Fig. 8.22) looks at the current position and the target position, and makes an appropriate signal to move the motor. Experienced motor controller manufacturers will use some moderate level strategies to use the feedback intelligently to achieve optimal performance. This is a rich topic for a good book on motion control theory.

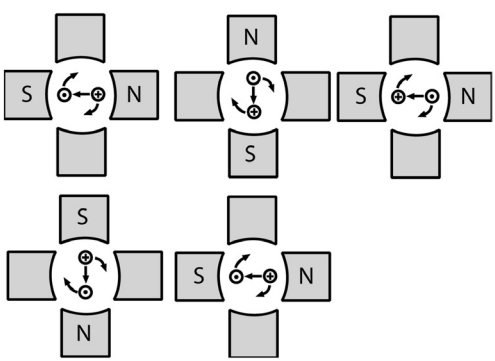


FIG. 8.21 Rotating stator field

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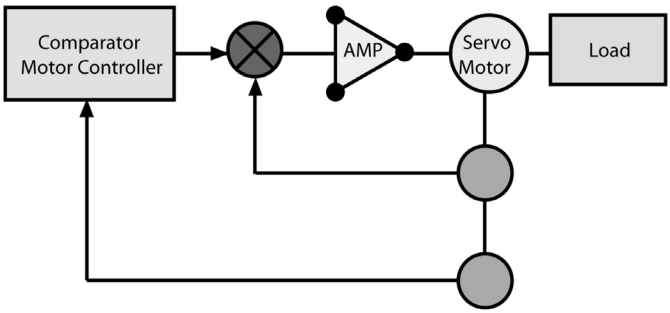


FIG. 8.22 Servomotor drive block diagram

So if one dictates a desired final position, speed, and acceleration, one can get a reasonably close output trajectory that will usually suffice for most applications. Servomotors and their supporting electronics are not outlandish in costs, but are not cheap either.

8.5.2. Electric Stepper Motor

Many applications of motors, not limited to automation, want to control an electric motor and hold it at a fixed position. Products like the dot matrix printer helped to drive the electric stepper motor technology so that very low-cost control could be achieved. Advanced stepper motor technology created significantly larger motors, many capable of moderate tasks within automation. The basic stepper motor shown in Fig. 8.23 has two sets of windings, and a permanent

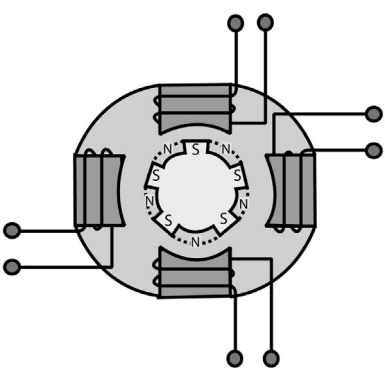


FIG. 8.23 Permanent magnet stepper motor

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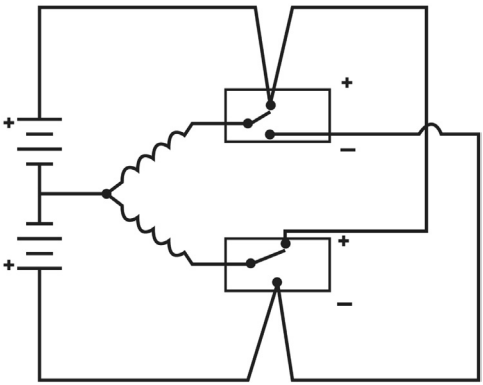


FIG. 8.24 Stepper motor circuit

set of magnets connected to what looks like teeth labeled "N" and "S" for North and South on the rotating center core or rotor.

By systematic switching of the windings (Fig. 8.24), the N and S teeth of the rotor are propelled or repelled to create a repeatable rotational motion. The motion can be in either direction, and can be controlled (Fig. 8.25) for speed and acceleration. The original single step motion process was first augmented to a half step process, and now uses micro stepping. The jerky motion of original dot matrix printers can now be as smooth as a finely tuned servomotor.

Stepper motors can work either with open loop control (no feedback) or closed loop (feedback from a rotary position sensor). The largest application base is in the open loop mode, since products like current ink jet printers are highly cost competitive (and sold for a loss to get you to buy their ink!). Perhaps the only real limitation to address in open loop control is the need for the motor to "home" itself

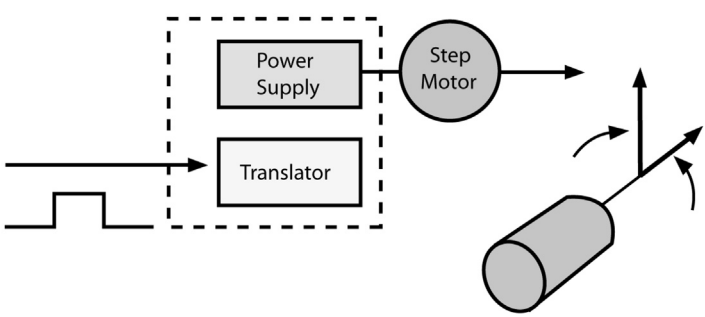


FIG. 8.25 Stepper motor with translator control

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by creating a contact with a limit switch or optical sensor. Without this information, the motor is only moving relative to where it currently is. So it needs a reference point to have an accurate position. Stepper motors, if over loaded, will not cycle their rotor's teeth properly, and behave in a manner referred to as "slipping."

8.5.3. Motor Sizing

Determining the correct motor size can be straightforward in many cases, and other times a challenge. The key concept with most electrical motors, whether DC continuous speed, electric servo, or electric stepper, is that there is some trade-off between torque and speed. Figure 8.26 shows a very generic diagram that has a linear relationship between the two. For most motors it is far from linear. One should always investigate the actual torque speed curve supplied by a motor manufacturer before one ever starts the purchasing process. These curves are usually quite accurate, and are the primary information used for sizing.

Generally, an application's torque and speed want to be under and to the left of the curve. Where one is in this region is open to engineering debate. In comparing similar types of motors but of different sizes and power, one would look at:

- What is the correct size (torque–speed) motor available?
- What is the impact of the larger motor's size and weight?
- What sizes are the other motors on the machine?
- Will the motor stall out?

Let us look at each issue.

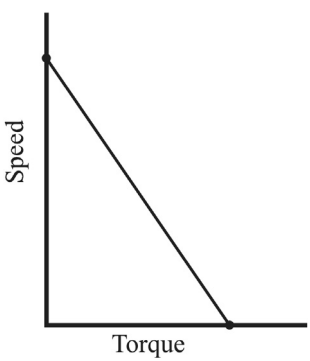


FIG. 8.26 Motor torque vs. speed

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8.5.4. Motor Selection

A stepper motor could have a torque–speed curve that looks something like Fig. 8.27. This curve is showing a continuous duty cycle, where the motor is being used a high percentage of the time 24/7. If the motor was being operated only minutes per hour (called intermittent duty) it could be run at higher speeds or torques. This would usually be designated by a second curve on the same diagram, or an entirely separate diagram. This is analogous to a human muscle being able to do a greater task for a short duration than what can be done for an eight-hour shift.

An "X" has been placed at what could be a good operating point. There is nothing special about this X, other than it is not very close to the speed or torque limit. Since most automation is being designed from scratch, the actual output of the motor is not always crystal clear. So it is possible one selects a motor with a fair factor of safety and the cost impact is not significant compared to changing the motor months later when debugging. If this motor application was to be duplicated by the thousands, then one would want to optimize as much as possible to get the most out of the cheapest motor available.

Motors are being optimized to new levels by some of the European robot manufacturers. Since most robot makers do not manufacture their own motors, they buy the same motors as their competition. So if one producer wants to get higher performance, they will run a motor on the brink of trouble. They will run the motor so it gets hot and is seemingly ready to fail, but it does not. This emboldened strategy takes some guts to pull off, something the author does not recommend easily!

Motor's Weight and Inertia

Robotics is a good example of why not to select oversized motors. The basic industrial robot arm design has the motor which drives any particular link situated

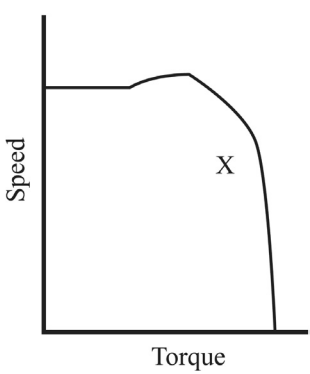


FIG. 8.27 Stepper motor torque vs. speed curve

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adjoining the link. So the motor closest to the robot's base must rotate all of the robot links, the payload, and the other links' motors. And then if the loads get bigger, the links must get larger to avoid deflections. You now have a vicious cycle of a robot weighting 2000 lb that can only carry 5 lb. In recent practice, a 2000 lb robot might carry 100 lb accurately, again not as efficient as a human, percentage wise.

In smaller applications, the rotating inertia of the motor's core is significant with respect to the rest of the mass to be moved. A quick analytical check is good to see if this will have impact, or is just a few percent. All of these comments assume that the motor connects directly to the output shaft. There is often a gearbox of some sorts, so the inertia the motor sees needs to be calculated on a level playing field.

Sizes of Other Motors on the Machine

Sometimes when all of the multiple motors on a machine have about the same torque–speed requirements, it is logical to select a single motor size for them all. The downside is that for one or two motors you have power overkill and extra costs, but for spare parts and support reasons the customer might be happier knowing that if they keep one extra motor on hand, their technicians can replace it in a matter of hours rather than wait for a service call. This can be an excellent reason to keep with one motor size, but it can also be an excuse for a lazy engineer to not bother with the sizing at all, and to just continue to use a favorite motor, that maybe happens to be sitting in inventory on the builder's shelf.

Stall Torque

This is a term that may be foreign to many, so one might be asking:

- What is it?
- Do you want to use it?

The best example of stall torque is using a battery operated power screwdriver. As one tries to twist a screw into a board that has a knot hole in it, the screwdriver will likely grind to a halt before the screw is completely inserted. The motor has stalled out, and the output torque has reached its peak, although not enough to do the complete job. The amount of this torque at the screwdriver's last gasp of life is the stall torque. It is that lower rightmost point on the curve in Figs 8.26 and 8.27.

Does a machine builder want to approach the stall torque? Most likely they do not. Similar to the battery screwdriver, one is creating the largest energy drain and mechanical strain on the motor system. This will produce a result that will likely be unpredictable. Imagine judging the results of how far a screw twisted into one knot hole, and assuming that at the next knot hole there will be similar results. Anyone who has worked with wood will know that this is improbable at best.

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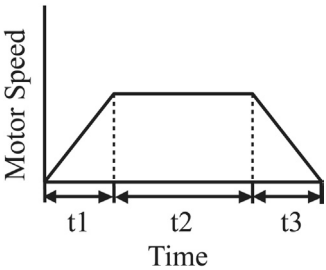


FIG. 8.28 Motor speed vs. time profile

8.5.5. Motion Profiles

A very large body of research has been performed to better understand motors and the effects different motion profiles have on actual performance. Figure 8.28 shows a common motion profile. The motor has an initial time segment where it is ramped up to a desired speed. The derivative of this curve implies a constant acceleration. Then the motor keeps a constant speed for a majority of the motion, followed by a deceleration or ramp down. This ideally should be easy for either a servo or stepper motor, but if it is an open loop control stepper motor and the ramp up or down is excessive, the motor can slip and the performance becomes unsatisfactory.

Assuming a gearbox of some type is used, the motion profile of the motor will match the profile of the application. Figure 8.29 shows a table that is to move, and its speed vs. time. A good point to note here is the reverse problem, which is usually what one sees when sizing a motor, and which is where the table speed has requirements for seemingly instantaneous acceleration and deceleration to and from a constant speed. This can be done successfully, but implies a much larger motor to reduce the ramping time. This will also drive up the costs. So the automation builder is wise to double check that the application really

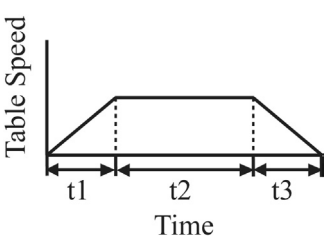


FIG. 8.29 Table speed vs. time profile

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needs such fast acceleration (and larger motor) before designing the machine. Perhaps the customer does not know what their requirements are implying.

Advances in Electric Motors

There are many advances in electric motors and their control. With many markets using motors far beyond the limited realm of automation, the competition to develop better products is present. This includes:

- Stepper motors packaged with threaded shaft drives to produce linear motion;
- Linear stepper motors;
- Rare earth motors and other improved materials.

The first two choices give more off-the-shelf options to automation builders. The first saves considerable time and effort of designing and building a unit that has often been built from scratch in the past. The rotational stepper motor is a known quantity, and its motion is coupled via a timing belt to a threaded shaft. The shaft can then extend or contract in proportion to the stepper's rotation.

The second choice is even more compact. It takes the North and South teeth of the stepper motor rotor, and in a sense, unrolls them into a linear bar. The energizing coils are also unrolled, and thus the linear toothed bar can step to the left or right. The bar can be connected as a motion output rod.

Strengths and Weaknesses

The strengths and weaknesses of the various arrangements are shown in Table 8.1.

8.6. AMPLIFIERS, DRIVERS, AND TUNING

The available signals out of most controllers are either digital or analog, but at a 5 V level. The 5 V amperage is at the logic level that the processor board runs at, and at best can light up an LED on a circuit board. It will not be strong enough to run a 5 V

TABLE 8.1 Strengths and Weaknesses of Various Arrangements

	Pneumatics	Hydraulics	Servomotor	Stepper motor
Performance	Good	Good	Great	Great
Speeds	Fast	Fast	Fast	Moderate
Loads	Moderate	Heavy	Moderate–heavy	Moderate–light
Cleanliness	Needs filters	Messy	Good	Good
Noise	Banging noise	Pumping noise	High pitched	Moderate
Costs	Low	High	Moderate	Low

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DC electrical motor. So the signal needs to be amplified, just like a portable CD player needs to be amplified before driving one' surround sound speakers.

There are several choices available to an automation designer, depending on the brand and type of electric motor to power. One can find that:

- Controller's analog voltage gets amplified to power the motor.
- Controller's digital signal gets converted by an amplifier to power the motor.
- Controller's digital signal instructs a smart amplifier (driver) to power the motor.

The first two are logical expectations, taking a signal and boosting it to make a large actuator move. There is usually an integral power supply or else an external power supply is needed. The third option allows for the controller to instruct a dedicated microprocessor (driver chip) to execute a trajectory sequence that the programmer can select beforehand. Any sensors to supply feedback send their signals to the driver chip, and the controller is only told by the driver that either the sequence was properly executed, or that there was an error.

Amplifiers have various levels that need to be set, the most standard being the gain. This is the multiplication factor by which the input is magnified. Some amplifiers have their settings adjusted by small set screws, so human setting of the gain with a small screwdriver is sometimes a guess since there is no feedback of the value being chosen. Trying to get multiple amplifiers to have the same setting is not trivial with this feature. Other amplifiers use computer software to download gain values, and this method is much more repeatable.

As will be shown in Case Study No. 2, there are also amplifiers that are self tuning; that is, they determine the gain and other parameter setting by exercising the actual motor connected to the actual load. The internal program may run anywhere from a few seconds to several minutes, putting the system through its paces. This can be a useful tool for creating new electromechanical systems, but it is not foolproof. The author had a system that was never tuned successfully, even though the product representative ran the self-tuning program 25–30 times.

A new approach performed in the laboratory has been developed by Dr. John Rogers (Rogers and Craig, 2003). This work uses open loop stepper motor control, and uses both computer modeling and actual hardware testing to determine the optimal motor stepping sequence. This sequence can then be repeated in multiple manufactured products like ink jet printers to achieve optimal performance without sensory feedback. As long as the products are similar in mass, friction, and motor strength, they will perform well. If a printer has excessive friction due to a manufacturing flaw, it will not, however, compensate for this deviation.

For all types of amplifiers and power supplies, it is common to place all of these and other electrical devices in a metal control panel. This panel can reside on the side of an automatic machine, or it can be mounted on the nearby wall. It

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also usually contains the circuit breakers or fuses for the line voltage for the machine, and has a primary switch that can be locked out for service and repair.

8.7. CASE STUDY NUMBER 1: STEPPER MOTOR SIZING*

In Chapter 11 there are detailed discussions about several new automation machines, one of them called the Trackbot. The overall application market and design goals will be left for that chapter. An integral subsection of the Trackbot will be presented here as an example for stepper motor sizing. This sizing effort was a challenging one that will not be totally finished until the next generation Trackbot is complete, but the lessons learned to date are worth noting.

The mobile part of the Trackbot is called a "Bot." Multiple Bots (2 to 12) travel around a closed loop track, come to a complete stop, and pick up a pouch from a conveyor. The Bot's pickup motion is the telescoping downward of a cylinder that has a suction cup on the end. The cylinder is contained within a larger tapered cylinder. This section of a Bot without its electronics, pneumatics, and cover is shown in Fig. 8.30. At the top is an electric stepper motor, which is connected to a mechanical housing, itself connected to the tapered external



FIG. 8.30 Bot with stepper motor, external body, and suction cup

*This case study supplied by Mr. David Brown.

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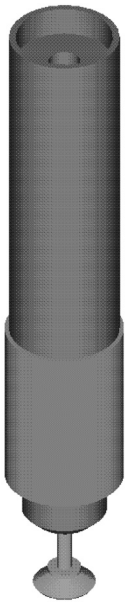


FIG. 8.31 Bot without external body or motor

cylinder. A small section of the telescoping cylinder with the suction cup attached is visible on the bottom. The telescoping cylinder and suction sup can be lowered 6 in. from this home position.

The telescoping cylinder and suction cup are shown as a separate unit in Fig. 8.31. There is a band around the lower section of the cylinder made of a low-friction plastic. This is the sliding bearing surface that facilitates smooth motion with the external tapered cylinder. The hole in the top of the round cylinder is for a threaded shaft.

Figure 8.32 shows the round cylinder with the threaded shaft, although the threads are modeled as a rod in this figure. At the top of the shaft is an attachment plate that connects the shaft to the small cylindrical helical coupling. This small coupling (which was the weakness of the mechanical design) is connected to the stepper motor, and allows for slight misalignments to be forgiven. The supporting framework has been removed from the figure for clarity. Figure 8.33 focuses on the threaded shaft and the flat round nut that connects the cylinder (not shown) to the shaft.

This mechanical and electrical hardware setup represents many similar applications, but its implementation on the Bot has some unique requirements:

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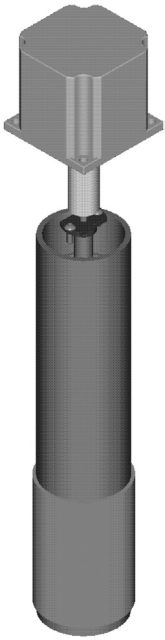


FIG. 8.32 Stepper motor connected to telescoping cylinder

- The Bots are moving on a track that has two electrical rails. The Bot has roller contacts, similar to many commuter trains, to power all electronics and motors. There are no batteries on the Bot itself.
- The Bot is to lift a pouch with its suction cup that weighs up to 1 lb.
- The cycle time is 0.25 sec to travel either downward or upward, so the speeds are demanding.
- There is a significant size limitation for the stepper motor amplifier/driver.
- There is no position sensor, so the stepper will work in open loop control.
- Motion profile will be generated by a single board PC on the Bot.

The demanding cycle time implies that a normal threaded shaft would have to spin at extremely high speeds in order to move fast enough. Imaging how fast one would have to turn a nut to get it to move 6 in. along a standard bolt in 0.25 sec. There were no stepper motors that could turn this fast available. So a very aggressive threaded shaft and nut were used. Its pitch was for 1 in. of travel

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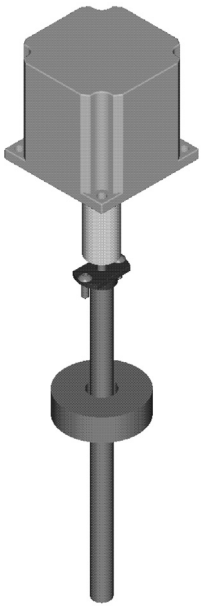


FIG. 8.33 Stepper motor connected to threaded shaft via the helical coupling

for each revolution. You would not usually even refer to this as a threaded shaft compared to the standard bolt.

The track's electrical rails were earlier determined to be 12 V DC, due to some other constraints. The best stepper motor choices were in the 24 V operational range. So a step-up transformer was placed on each Bot. There were no step-up transformers for much higher voltages available in the year 2000 that would fit on the Bot.

The system was built and debugged for months. The stepper motor performance was erratic. The telescoping motion would appear to freeze up on random downward and upward motion. The stepper motor was slipping from its being pulsed too aggressively. The randomness came from environmental conditions, such as room temperature and the relative friction of the tapered cylinder on the round cylinder's low friction material band. Also, the nut on the very aggressive threaded shaft would sometimes stick rather than move smoothly, even though the nut was made from low-friction plastic.

There were other performance issues that clouded the debugging picture. One was that the Bot's internal connectors were not of high enough quality for the stopping and starting motion. The connectors used on this first prototype were standard electrical connectors use in PCs and similar stationary electronics.

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Loose connections happened way too often. One was never entirely sure which problem was causing a stepper motor failure when it occurred. Automotive type connectors are being used on the next generation Trackbot.

The time and motion constraints meant that the motion profile (Fig. 8.28) was being taxed. If the ramping up and down from the constant speed was too quick (causing the stepper to slip under load) then a longer ramp time meant that the constant speed portion needed to be at a faster rate in order to get to the end goal in time. It also did matter if the motion was up or down due to gravitational effects, and it mattered if the suction cup got a 1 lb pouch or came up empty.

Different ramping curves were tried, including a cycloidal motion (Downie, 2000). This motion was generated within the single board PC on the Bot, and did help to improve things, but not enough to get 99.999% performance. With the motor and amplifier system used, there was no self-tuning process that could have been used either.

As of 2004, there are several recommendations to try to improve the situation:

- Upgrade the rail power to 24 V, allowing for stronger stepper motors. If 24 V alone is not sufficient, use the existing step-up transformer to get 48 V. Or if 48 V is not enough, combine the 48 V with the 24 V to get 70 V.
- Smaller amplifiers/drivers that are more powerful are now available compared to 2000.
- Better amplifiers/drivers with built-in speed ramping options are now available.
- Investigate present-day electric servomotor systems.
- Use pneumatics if sliding positional accuracy not critical.

8.8. CASE STUDY NUMBER 2: SERVOMOTOR SIZING*

A United States based company designs and makes custom safety covers for swimming pools. While the company makes regular stock safety covers that fit standard-sized pools, their main product is the custom safety cover. Some homeowners build oddly shaped pools. Since these pools are not any standard size or shape, they cannot be covered with a stock product. Instead, these pools must have covers tailored specifically for them.

A typical cover is made of several pieces of material. Since covering a swimming pool with one large piece of cloth is infeasible, the cover must be designed from several narrow (3–6 ft in width) panels of cloth stitched together and reinforced by 1 in. wide strips of cloth webbing. Each panel has a unique

*This case study supplied by Mr. Peter Caratzas.

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shape that, when stitched together, creates a cover that approximates an oversized outline of the swimming pool. In addition to support of the panel seams, the webbing also provides a means of mounting the cover to the decking around the pool. Buckles and hooks attached to the reinforcement webbing connect to anchors sunk into the decking surrounding the pool.

The process of making a custom safety cover begins with the measurement of the customer's pool. Once all critical dimensions are obtained, the information is faxed to the design department. Using AutoCAD, along with several functions developed especially to automate the design process, the draftsperson generates a plot of the finished cover, which states the amount of each type of material needed.

The completed design is released to the manufacturing floor. Here, according to the plots, workers called cutters draw out the general outline of each panel, the location of each reinforcement strap and other features necessary to manufacture the cover. General outlines of the cover are marked in yellow crayon and attachment information such as the locations of reinforcement straps and buckles are marked in white crayon. The transposition of information from the plot to the cloth is the most error-prone step in the entire manufacturing process. To attain a 100% quality of the output, this stage requires the cutter to maintain a very high level of concentration during the entire day — an almost impossible demand. In addition, this step requires a great deal of knowledge and experience because other factors that will determine the quality of the finished cover are set at this step.

The material used to manufacture the cloth is an open-weave polypropylene cloth. There is a good deal of variance in the quality of this cloth. In fact, the manufacturing process is so prone to flaws that it is rare to find a roll of material that is longer than 150 yards that does not contain at least one flaw. It is common for the loom to skip a thread, leaving a gap in the material equal to one thread length in thickness. This is a problem called a line flaw. This is a noncritical flaw that does not weaken the strength of the cover, but does affect its aesthetics. Knowing where the flaw is, the cutters can plan to locate a reinforcement strap over it, hiding the unsightly line flaw from the customer while reducing the amount of wasted material. A second flaw type is an area flaw, which is a defect in the material than needs to be avoided.

The manufacturer of the cloth inspects each roll of its product for the various types of flaws. Inspectors mark the area flaws with strips of yellow adhesive tape and line flaws with strips of red adhesive tape.

8.8.1. Proposed Improvements

The ideal solution would incorporate a system that would maintain the in-process quality checks that guarantee a quality product coupled with a manufacturing system that ensures predictable throughput. After studying the process, it was

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determined that the greatest area for the introduction of flaws was through human error. These errors were most likely to occur in the marking and cutting operation and had the greatest impact on the process throughput.

The RPI Center for Automation Technology (CAT) developed a machine that takes the human decision making out of the marking process (Caratzas, 2002). By eliminating the operator from interpretation of the design plan, the number of errors introduced in the marking process is reduced significantly. Instead, the operator runs a machine that interprets the design data and generates the design.

8.8.2. Hardware Requirements

Since the cloth contains flaws that cannot appear in the finished cover, and these flaws are marked by colored tape while the cloth is being manufactured, some sort of sensor had to be developed that could find these pieces of tape as the cloth was being pulled out. This sensor would have to signal the motion controller to stop the material outfeed and give the operator options to eliminate the flaw in the material.

The manufacturing facility is a light-industrial site. It does not have pneumatic or hydraulic lines installed, and only a very few 220 V circuits to run the large "Union-Special" style sewing machines. It was hoped that the CAT could supply a machine that could run at 110 V with minimal pneumatics or hydraulics so that minimal site preparation would be required.

The most important requirement to the company was the one that would justify their investment in the project. The completed machine would have to mark the cloth panels more accurately and quickly than the manual cutting method did. If the designers released designs that were correct, the machine should definitely make fewer mistakes, if any at all. Throughput was the second part of the requirement that would have to be met if the project was going to be a success. The company estimated that their cutters needed 15–25 minutes to lay out all the yellow and white lines on all the panels that make up one cover. The ten-minute spread in time was dependent on the complexity of the cover and number of panels in the completed cover. The company wanted to minimize this time. In the CAT's discussions with them, it was decided that they would be satisfied if the throughput time could be reduced to a range of 8–15 minutes per cover. After measuring the length of each line and estimating the speed at which the machine could mark, the machine's time to mark a cover was determined. This was added to the time required to pull the raw material out onto the table and remove the finished cloth from the table. The time required to mark the sample cover was estimated to be 9 minutes. To achieve this goal, the machine would have to mark at a speed of 24 in. per second and pull out material at a "rapid traverse speed" of 48 in. per second. In addition, when

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the machine was moving but not marking, as it might when moving from the end point of one line to the beginning of the next, it would operate at the "rapid traverse speed."

8.8.3. Basic Overall Concept

The company wanted this machine to be as similar in function to the manual method of marking as was possible for two reasons. First, space in their facility is limited. If the machine ever broke down for an extended period of time, they wanted to be able to use the space that the machine occupied for manual marking and cutting of covers. Secondly, they wanted the cutters to make transition from manual marking to automated marking with as few problems as possible. It was felt that if the machine mimicked the tasks that the markers performed, the workers could easily identify which task was being performed at each stage and monitor that the machine was operating properly. By reducing the learning curve associated with operating the machine, the company could realize benefits of the device quickly.

The basic design is analogous to a large flatbed $x-y$ plotter. A photograph of this is shown in Fig. 8.34. In this case, the bed is a 64 ft long and 7 ft wide table. The bed can accommodate the 6 ft wide cloth that is used to make the covers. A gantry straddles the 7 ft width and is designed to move back and forth along the length of the table. A linear slide mounted to the cross member



FIG. 8.34 Basic machine overview

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of the gantry spans the width of the table. A carriage on the slide holds two marking heads (pens) that spray the yellow and white ink onto the material. This configuration gives the carriage two degrees-of-freedom and allows it to move anywhere within the 64 ft by 7 ft work envelope. Each marking head is mounted to the carriage via a pneumatic cylinder. The cylinders allow each head to extend to the proper standoff distance when they are marking and retract when not in use. The carriage also holds two ink reservoirs, each of which holds eight ounces of ink, yellow and white respectively. There is a cable chain attached between the table and the gantry and one attached between the gantry and the carriage. Any cabling or pneumatic hosing passes through this cable chain to prevent their tangling and snagging as the carriage moves through its full range of motion. Solenoid valves are mounted to the gantry to control the actuation of the cylinders that raise and lower the heads and to activate the spray heads. There is also a flashing yellow safety beacon and warning beeper attached to the carriage that are activated whenever the gantry is in operation to warn anyone in the vicinity that the machine could move at any time.

The Drive System for the *X*/*Y* Axes

A variety of methods were investigated to drive both gantry and the carriage, and a variety of factors involved in selecting the drive scheme for each axis. Most important was accuracy. The company stated that the machine must be accurate to the nearest $\frac{1}{8}$ in. over the entire length of the table. Given the length that the gantry had to move, a chain drive or cable drive looked the most promising initially. Cable drive was almost immediately eliminated because there was concern that stretch in the cable would lead to inaccuracies. In addition, the cable would have to be checked periodically for fraying. These maintenance issues quickly eliminated a cable drive system.

The second method investigated was a chain drive. Once again this showed initial promise because of the ease of implementation. It would be relatively simple to purchase 128 ft of chain, join it in a loop and drive it from one end to get the gantry to move back and forth. Concern quickly arose about the accuracy of such a drive system. Given the mass of the gantry (95 kg), the design team questioned if the chain would not stretch as the cable would. In addition, the chain would add mass to the system being moved, meaning that larger motors would be required to drive the gantry. Finally, since the chain is flexible, there was concern that this lack of stiffness would adversely affect the speed of response of the system. Given the chain's tendency to bunch when pushed and stretch when pulled, it was felt that it would be difficult to predict the dynamics of the chain drive during acceleration and deceleration.

For the *Y*-axis (carriage), a lead screw drive was also investigated. Since the carriage only had to move over a length of 7 ft, the design team thought that some

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sort of lead screw or ball screw would be an alternative. With this setup, there would be no reason to worry about the dynamics of the system. The lead screw is stiff along the axis of motion, and its behavior over the motion profile is consistent. This system was eliminated for another reason. The rotational velocity was calculated to find how fast the *Y*-axis motor would have to spin to get the carriage to move at the specified linear velocity. For a typical helical lead screw, the rotational velocity was fast, but attainable by most servomotors. The concern was that given the spinning a 7 ft long lead screw at the angular velocities that were required might cause the lead screw to start whipping. Whipping could lead to premature failure of the lead screw's nut and inaccuracies in positioning.

The final method investigated was the rack and pinion drive scheme. The design called for gear racks to run up each side of the entire length of the table. Meshing pinions would be attached to shafts at each end of the gantry. These shafts would be connected to the *X*-axis drive motor by a gearbox that would split the input torque of the motor shaft into output torque at two separate shafts. By driving both ends of the 7 ft long gantry, the moments driving the gantry would be balanced and it would track straight along the table. Since this scheme is a direct-drive method, it was felt that positioning would be most accurate with this method. The only uncertainty in position would be the result of backlash in the gear train. The backlash could be controlled in two ways: first, by purchasing gear heads with negligible backlash; secondly, by designing adjustment into the shaft hangers that held the pinion gears in mesh with the rack. Adjusting the distance between the rack's pitch diameter and the pinion's pitch diameter could reduce backlash between the rack and pinion. The rack and pinion system provided the greatest benefits in our configuration and was chosen for both *X*- and *Y*-axes.

Unspooling Schemes

Developing the drive system completed one major portion of the conceptual design — accurately controlling the gantry as it traversed the workspace. There were other functions that the final design had to perform. The machine had to be able to autonomously spool raw material off a roll and pull it to an optimal length for the given panel it was making. The material would have to be clamped in position during the marking stage and then released when marking was completed. Finally, the finished cover had to be cut from the roll of raw material to which it was attached.

In deciding how to pull out the material, two schemes were explored. Each was developed around a bar that ran the width of the table and had clamps attached at points along it. The bar would be supported on wheels that would allow it to roll up and down the table. In the first concept, this bar would not

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be powered. This bar would be driven by the moving gantry. In order to pull material out, this bar would dock to the underside of the gantry. Once the bar's clamps were fed material and they had tightened on it, the gantry would pull the clamp bar and its payload to the desired length. When this bar had reached it destination, it would release itself from the gantry and, via brakes designed to attach the bar to the racks running up each side of the table, lock itself to the table. When marking was complete and the finished panel was removed from the table, the sequence would be performed in reverse, and the moving clamp bar would be brought to a home position to wait the next feeding of raw material.

There were disadvantages to this system. First the gantry had to make at least one extra trip up the table to either drop off or pick up the moving clamp bar. This trip would add to the cycle time estimates to complete the cover. In addition, since the moving clamp bar itself would add mass to the system, it was not clear how the tuned motion controller would behave when the added payload was attached to the gantry.

In the competing design, the moving clamp bar gantry would be self-propelled, driven by a third axis on the motion controller. There were several advantages to this system. A lightweight moving clamp bar could traverse the length of the table more quickly than could the gantry. Also, the gantry would not have to make the extra trip down the table that was required by the unpowered clamp bar. Pullout speeds would be faster and machine cycle times would be lower. The disadvantage to this setup was that a third servo axis on the controller would be required to implement this design. The added axis, while being used infrequently, would add considerable cost to the control system. In addition, the added motor drive would add more backlash adjustments and overall maintenance requirements to the system. After looking further into the amount of time that would be added by the gantry's extra trip down the table, it was decided that the benefits of the powered clamp bar did not outweigh the costs, and the unpowered clamp bar was chosen.

8.8.4. Actuator Sizing

Table 8.2 is a summary of the system as installed at the company. It gives the machine dimensions and specifications. Major hardware components of the drive system for both axes are also given.

The design of a motion control system (Fig. 8.35) is an iterative process. The system will include a motor coupled to a load. The coupling may take one of two forms: direct-drive or speed reduction. In direct-drive, the system is coupled directly to the motor. This may reduce the efficiency of the system, but increases it's accuracy, since the load will be directly coupled to the encoder, as well as the motor.

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TABLE 8.2 General Machine Specifications

Overall dimensions (l × w)	64 ft × 7 ft
Maximum width of cloth	63 in.
Usable length of table	61 ft
Weight of gantry	205 lbs
Marking speed	50 in./sec
Pullout speed of material	50 in./sec
Number of inputs	6
Number of outputs	12
Controller	Galil DMC-720
X-axis information:	
Motor	Galil Nema 34
Gear reduction	Bayside 5 : 1 Planetary
Right angle gear drives	Zero-Max 5/8″ Shaft
Pinion	Browning 3/4″ face, 16 pitch, 24 tooth
Rack	Browning 3/4″ face, 16 pitch
Couplings	Lovejoy Spider 5/8″ i.d.
Amplifier	AMC 25A12
Power supply	AMC 80 V
Y-axis information:	
Motor	Galil Nema 23
Gear reduction	Bayside 5 : 1 Planetary
Right angle gear drives	N/A
Pinion	Browning 1/2″ face, 24 pitch, 24 tooth
Rack	Browning 1/2″ face, 24 pitch
Couplings	N/A
Amplifier	Galil Amp 71
Power supply	Integral to amplifier

On the other hand, to increase system efficiency, the load can be connected to the motor through a speed reducer. Speed reduction can be accomplished by gear reduction or pulleys. Speed reduction will reduce the size of the motor needed to drive the system, but may introduce inaccuracies into the system through backlash in the gearing components. The designer must determine the critical criterion.

The first step performed in sizing the motion control system of the machine was the determination of the system load. This was determined by modeling the mass of the gantry. Masses for each component were estimated and the total gantry system was determined. The equipment was proposed to run at a certain speed, and had to accelerate to speed from a stop. The maximum

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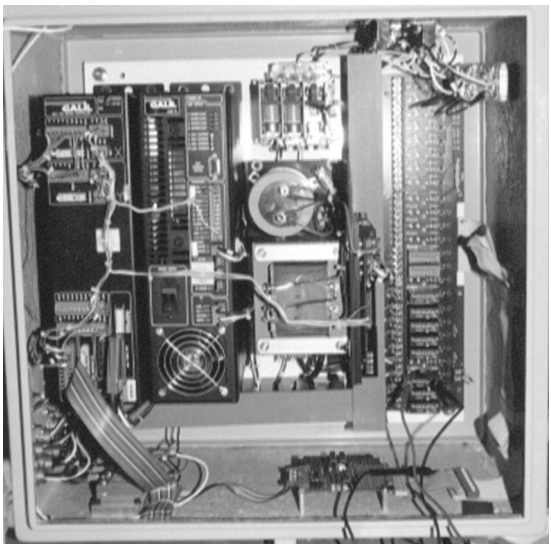


FIG. 8.35 Control cabinet including all motion control components

acceleration was determined so that the machine could reach the proposed average velocity. With the system mass and acceleration known, the force required for the system was calculated. Once the force was known, the torque could be determined. Since the load was driven by a rack and pinion drive, the torque was calculated using:

$$\tau = r \times F \tag{8.1}$$

where τ is required system torque, r is the radius of the pinion gear, and F is the force required by the system.

At this point the process became iterative. The investigation was performed to determine the need for gear reduction. As previously stated, gear reduction allows the designer to select the most efficient motor for the load being driven. In other words, the designer can select the smallest and, most likely, cheapest motor for the application. Since torque is increased as a multiple of the gear reduction at the output shaft of the gear reducer, speed must be reduced by the same factor. Care must be exercised to ensure that the motor will be able to meet the speed requirements of the system if gear reduction is used.

To begin the analysis of whether or not gear reduction is needed, a motor was selected that met the necessary torque requirements. The rotor on the motor has an inherent moment of inertia, J_M. An optimal system will have J_M equal to

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the moment of inertia of the system, J_{LO}. If not, gear reduction must be used. For the system, it was determined that gear reduction was necessary.

If gear reduction is used, the inertia matching must be performed by comparing J_M to the reflected load of the system at the input shaft of the gearbox, J_L. In this case, the comparison takes the following form:

$$J_L = \frac{J_{LO}}{N^2} = J_M \tag{8.2}$$

where N is the gear ratio of the gearbox. Rearranging terms, it was found that the optimal gear ratio was:

$$N = \left(\frac{J_{LO}}{J_M}\right)^{1/2} \tag{8.3}$$

As a rule of thumb, J_{LO} should be less than ten times J_M.

8.8.5. Sizing the Amplifier

The amplifier drives the motor by providing power to coils in the rotor. Since $P = VI$, the two factors that affect motor performance will be current and voltage. The peak torque that can be generated by the motor is a result of the peak current provided by the amplifier. Motor manufacturers will publish a torque constant (K_T), which relates torque to amps provided. To reach maximum designed acceleration, the motor must generate its maximum torque (T_P) and the amplifier must supply its maximum current. To determine the peak current required (I_{max}), the following equation was used:

$$I_{max} = \frac{T_P}{K_T} \tag{8.4}$$

Similarly, for the system to reach the maximum designed velocity ω_{max}, the amplifier must generate its maximum voltage (V_{max}). To determine the maximum voltage, the following equation is used:

$$V_{max} = K_T \omega_{max} + rI_{max} \tag{8.5}$$

where r is the resistance of the motor. The first term is the back EMF of the motor, and the second is the voltage required by the circuit created by the motor's coil. The importance of these numbers cannot be understated.

8.8.6. System Troubleshooting

As the gantry was installed on the table for the first time, it was subjected to manual testing. The gantry was supported by rubber wheels, and interconnected to the table by the gearing system. Each side of the gantry had a spur gear that

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interfaced with a gear rack that ran the length of the table. To ensure that the marking would be as accurate as possible, it was necessary to minimize the backlash in the drive gearing. Backlash for the gearing was set by minimizing the gap between the spur gears on the gantry and gear rack on the table. This adjustment was performed as the gantry remained stationary at one end of the table.

When the backlash adjustment was complete, the gantry was rolled through its full range of motion by hand. As this test was taking place, it was found that the gantry was easier to move in certain areas, and more difficult to move in other areas. It was quickly determined that the gear racks on each side of the table were not parallel. The distance between the two gear racks did not remain constant along the length of the table. As a result, as the spur gears on the gantry rolled along these different areas, the backlash in the gearing would change. In some areas, where the distance between the gear racks was greater, backlash was increased, and the gantry moved more freely. In other areas, this distance narrowed and the minimized backlash in the gearing system would cause binding that would nearly prohibit movement in the gantry system.

The difference in backlash in different areas of the table would cause two problems. First, repeatability throughout the system would not be equal, since there would be more play in the system in different areas of the table. Secondly, since the torque required to move the system to different positions was dependent upon the amount of backlash at the current position of the gantry, concern arose about the controller's ability to compensate for the variation in the required torque. Both problems could potentially lead to inaccuracies in the location and repeatability of the gantry, and ultimately, the marked lines that the machine was creating. This would have been unacceptable.

Because each of these racks was 65 ft long, it was expected that keeping them straight would be difficult, and that some adjustment would be necessary to ensure proper operation. Because this machine was created at one location and installed in another, developing a repeatable alignment procedure was equally important. The alignment procedure designed was straightforward and was easy to implement in the field.

System Tuning

With the last screw tightened and the final wire installed, the first-time machine builder may think that the equipment is ready for delivery. This is not the case. Several steps remain to be completed in the process after the final dust cover is installed on the machine. Once the machine is assembled, the system is ready for power-up. Before the first pool covers could be made, however, the drive system for each axis have to be tuned. When motion control systems are involved, as was the case on this machine, the controller must be optimized to match the dynamics of mechanical system it is controlling. With multi-axis control systems, where a

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controller drives multiple motors, each of which drives a separate mechanical system, the controller must be tuned for each axis.

Traditional motion control systems use a Proportional Integral Derivative (PID) controller to drive the motor to its commanded position. The PID controller's name is derived from the three control methodologies used to generate the command signal: proportional, integral, and derivative. The proportional controller generates a command signal whose strength is based upon the difference between the actual position and the commanded position. As the mechanical system gets closer to its commanded position, the proportional controller will reduce the strength of the command signal. When the system is at its desired location, the command signal should equal zero.

The proportional controller is inherently flawed. As the mechanical system reaches its commanded position and the command signal decreases, mechanical losses remain fairly constant. As a result, this controller will always leave the mechanical system with some error between desired and actual location. This is called a deadband. If this error is great enough to cause the controller to generate a command signal that will overcome the system's mechanical losses, the mechanical system will approach its targeted position. Otherwise, it will remain where it is.

The addition of an integral controller eliminates the deadband. This controller sums the difference between commanded and actual position over time. As time progresses and these individual errors are summed, the integral of the error increases. Consequently, the strength of the corrective action of the integral controller increases over time. When this error becomes great enough to generate a signal to overcome the mechanical losses, the system will move toward the commanded position.

Like the proportional controller, the integral controller has inadequacies. An integral controller is more likely to cause the system to oscillate. The design of the controller causes this. As the error signal increases, the strength of the corrective action of the integral controller is increasing. As the system moves toward its commanded position, the magnitude of the error is decreasing. However, it is still error, and when added to the previous error, the control signal continues to increase. In fact, the integral controller will not send a negative control signal until the system overshoots its commanded position, thus generating a negative difference between commanded and actual position.

To remedy the oscillatory behavior of integral controller, a derivative controller is added. The derivative controller analyzes the rate of change in the difference between the commanded and actual position. This controller has a tendency to slow the system down if it is near the commanded position and approaching it quickly.

It is not necessary to always use the PID controller. Depending upon the system design and required accuracy to perform command-following, it is possible to

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implement a proportional-only controller or a proportional-integral controller. Selecting appropriate gains for each of the individual controllers can optimize performance of the system. Each gain acts as a scalar dictating the magnitude of each controller's action. For example, by setting a high gain on the proportional controller and low gains on the integral and derivative controllers, the proportional controller will have a greater effect on the commanded signal.

Gains are used as a means to adjust, or tune, the controller to complement the mechanical system. Gains are not chosen arbitrarily. The mechanical system will have certain frequency response based on the arrangement of different mechanical elements such as springs, dampers, masses, and forces. There are two strategies for determining controller gains. One strategy involves using modeling methods that can be used to determine the appropriate gains for the system. These methods determine gains by modeling the mechanical system and the controller.

Rather than model a system, a second strategy for tuning is to determine the frequency of the actual system and determine the gains based on real information. The advantage to this method is that it eliminates modeling error. The importance of this cannot be overstated. Modeling error can be significant, especially when trying to estimate such factors as frictional losses. Controller manufacturers generally provide tuning software with their motion controllers to aid the user in determining the appropriate controller gains.

Tuning was accomplished using software supplied by the manufacturer of the controller. The software determines the gains by applying step functions to the system. By measuring the response of the system to the step functions, a determination of optimal proportional, integral, and derivative controller gains can be performed.

When system tuning was completed, performance of the system could be verified. It was noticed that the covers were taking much longer to be completed than expected. Panels that were supposed to take two minutes to finish were taking eight minutes.

The motions that the gantry followed were analyzed to explain why the system was taking longer than expected to make the covers. It was determined that the acceleration for the system was set too low. Since the covers are comprised of many lines that are short in length, the gantry was never reaching its top speed. In this situation, the gantry's motion profile was triangular instead of trapezoidal — the desired profile.

To increase the throughput of the machine, the team attempted to increase the speed of the machine, both in marking and pullout. The first solution attempted was an increase in both the speed and acceleration of operation of the machine. This led to unstable behavior. The machine would overshoot wildly in the *X*-axis (the more massive of the two axes), and the position at which the lines were drawn would be inaccurate.

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In order to make sense of the behavior, it was decided to make a second-order model of the system. The shafts that drive pinion gears were modeled as springs and the spring constant of the system was determined experimentally. Friction in the wheels and at the interface between pinion and rack was arbitrarily approximated at a value of 10%. The gantry was weighed on a scale, and from that, the system mass was determined. Voltage and current outputs for the amplifier were known.

The model was implemented on Matlab and the results showed that the system was stable for the gains that were being used. This meant, inexplicably, that the system should be stable, yet it was not. The only information the team had not measured and therefore were unsure of had been the current output of the amplifier. When the team measured the current output to the motor leads, the team found that the amplifier was current-limited. The motor was not supplying the torque required or capable of, because it was not receiving enough current from the amplifier.

When the specifications of the amplifier were investigated in depth, it was found that although this motor and amplifier were sold as a matched set, the amplifier was incapable of providing enough current to get the peak performance out of the motor. In addition, the motor required 80 V to run at top speed, but the amplifier was only capable of supplying 50 V of maximum potential. This meant that the motor could not reach its top speed.

Since the Matlab model was already created, it was decided to add current limiting to the input of the system. When this was added to the Matlab program, the model predicted the observed behavior. By having this simulation available, the team was able to input the specifications of larger amplifiers/power supply combinations and determine which components would meet our needs.

Once the new amplifier was purchased and installed, the system behaved as predicted, with accuracy and speed. There was no overshoot and the system acted remarkably "stiffer." With these modifications, the system was installed in the factory and it has been in daily use since April 1995.

Curved Lines

Soon after the new amplifier was installed, and the machine was operating at the designed speed, a new problem revealed itself. Straight lines that required coordinated moves by the *X*- and *Y*-axes were being drawn as curves. During the initial debug, straight lines were drawn as expected. As the run hours increased on the machine, the "curved line" problem worsened.

There were two possible contributors to this problem. The problem could have been caused by the slippage in the mechanical drive system or corrupted feedback information. Corrupted feedback information could have been caused

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by a faulty encoder, noise in the feedback signal, or an incorrect translation factor between encoder counts and distanced traveled. For the lines to appear as curved, the problem had to be nonexistent in one axis vs. the other. If the severity of the problem was equal in both axes, the lines would appear as straight, since both axes would be moving with equivalent amounts of error.

Two steps were taken immediately to try and determine the problem. First, all conversion factors between encoder counts and distance traveled were re-checked to ensure that the controller was using accurate information for its trajectory generation. This scenario was the least likely to occur, but it was the easiest to check.

Mechanical slippage in the drive system of one of the two axes was the most likely cause of the problem and easy to check. All couplings and gears were tightened to their shafts. Lines were painted on the shafts and couplings to indicate where these parts were aligned. The machine was run again and, over time, similar behavior showed that the lines had moved in relation to each other.

To permanently remedy this situation, it was decided to cut keyways into the shafts, so they would be rigidly fixed to their couplings. The machine was run for several weeks and this change seemed to fix the problem. Slowly, however, the problem began to worsen. The painted lines on the shafts still indicated that shafts were not moving with respect to their couplings and inspection showed that the keys in the shafts were intact.

The two right angle gearboxes were never inspected for slippage when the curved line problem was found. They were selected to match the specifications of the motor. When the new slippage problems were encountered, the two gearboxes were dissected to look for potential failures. This inspection revealed interesting results. The gearboxes are designed with two shafts intersecting at 90° to each other. Beveled gears are rigidly mounted to each shaft. The interaction between the two gears transmits motion from one shaft to the other.

To prevent the gears from slipping, a hole is drilled through the gear and its mating shaft. A pin is pressed through this hole. When these gearboxes were dismantled, it was found that the pins had broken, and slippage was occurring between the shafts and the beveled gears. The question arose as to why the pins had broken, when the components were sized to match the motor. The pins were modeled as beams, with forces applied at the interface between the shaft and gear. Maximum torque was resolved into forces at these points and it was determined that, indeed, the pin should not break. This calculation was followed by cyclic loading calculations of the fatigue life of the pins, based upon fully reversed loading. Failures were revealed with this calculation. The vendor was contacted, and verified that these gearboxes were designed for continuous motion applications, not the reverse loading of a servomotor system. The application exceeded the design limits of the system.

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With the new data, two new right angle gearboxes were specified. The new gearboxes were designed specifically to meet the demands of a motion control system. The resulting replacement has been in service without fail since.

8.9. CONCLUSIONS

The correct sizing of actuators is needed if your new automation machine is to function properly. Simply using an oversized motor off your shelf can work sometimes, but other times come back to haunt you. To achieve correct sizing, one must calculate the loads and rely on the manufacturer-supplied diagrams and data sheets, but once in a while this information may be incorrect, so do not assume anything if performance is not what you expect.

PROBLEMS

1. Take a battery-operated screwdriver and create a stall torque condition twisting a screw into a very hard piece of wood. Devise a method to measure the stall torque with the aid of a spring scale or other device.
2. Using a battery-operated screwdriver, use your hands and the forward and reverse switches to attempt to create a very simple servo situation. Pick a direction corresponding to a map direction (South, Northwest) and try to servo the screwdriver blade until it comes to the correct direction. Mark the blade with a piece of tape as a pointer to help with this.
3. Borrow a battery-operated toy car and add extra loads to it, so as to create near stall torque condition (it can still move, but barely). Compare battery life to operation with the normal load.
4. Create a humanly operated vacuum cup system with a plastic straw and some other household items you can find (or go to the hardware store to be inspired). Determine how much you can lift by sucking on the straw if you have the open area of just the straw itself, and compare to some other larger opening by placing a larger opening on the end of the straw. (Be sure to use a stiff plastic straw and not paper.) Or use a larger straw in the form of a garden hose. From the weights lifted and the area of suction, calculate the negative pressure that you are able to achieve.

PROJECT ASSIGNMENT

1. Investigate the different actuator options for your project. Compare them and develop a trade-off matrix. Then select your first choice

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and develop the various subsystems. Keep your second choice handy and develop that choice to a lesser degree.

2. Search on the Web for leading vendors for your actuator choices. Look for their specific sizing information, and find one vendor who has a Web-based sizing program to run.
3. Estimate the total power needs of your actuators. For pneumatics, find the flow rate needed. For hydraulics, find the working pressure and flow rate. For electric motors, find the voltages and amperages.
4. List the number and type of inputs and outputs to power and control your actuators. What trade-offs can be made for fewer actuators and greater control signals?

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